

Boundaries and 'Not Sure'

Project: Helpful Classroom Sorter

1. Your mission

Your learning target

I can identify a specific source of uncertainty and explain how it affects a prediction.

I can compare predictions across stated possible inputs without inventing a missing value.

I can design a review step with relevant evidence and an explicit unresolved outcome.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

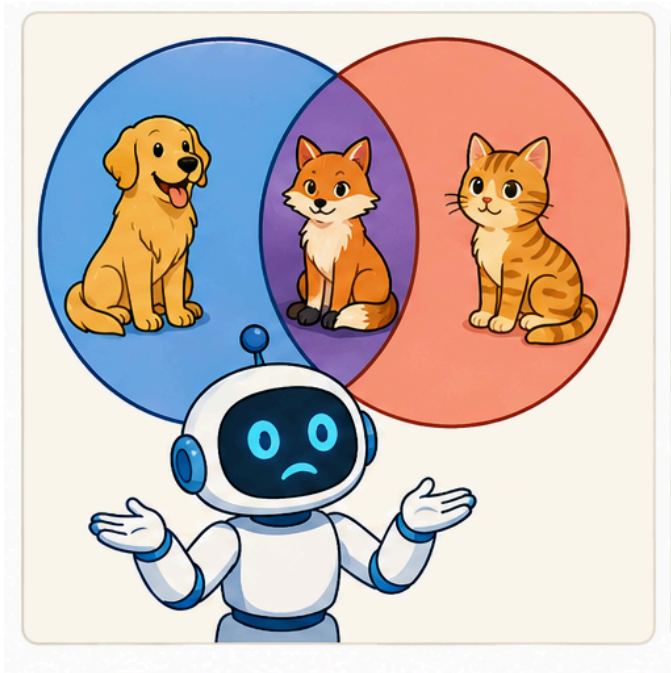
Without opening the old lesson, explain one key idea from Save New Examples for Testing. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Learning from Examples. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from If/Then Decisions. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Overlapping categories create uncertainty; abstaining can be responsible.



LEARNING TARGET A sorter receives a clear purple card, a red card hidden by a sleeve and a card labelled both red and blue by two reviewers. Give a different reason for uncertainty in each case and a suitable next action.

Learner name:

Readiness for the connected explanation

A model predicts tall at a height of at least 6 cm. What would it predict if an unclear record could mean either 5 or 6 cm? Does a good result on a small final test settle this particular input?

This week's end goal

By the end, I can use uncertainty to request a specific review when possible inputs lead to different decisions.

2. Notice the evidence

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; evidence tells us what the clues mean.



MYSTERY A soft toy shaped like a ball may fit toy, sports, or comfort-item bins depending on the sorter's purpose.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first prediction and state the clue that supports it.

My observations, prediction, and evidence:

GOOD THINKING A prediction can change when stronger evidence appears.

A closer explanation

An unclear measurement, a missing feature and a weakly tested model create different problems. Saying the AI is unsure does not identify which one matters. Write what is known, what remains unknown and how the unknown could change the result. Then choose a check that addresses that specific gap.

Use the familiar tower model: tall at a height of at least 6 cm, short otherwise, for whole-number heights from 1 through 10. A smudged card could read 5 cm or 6 cm. The threshold is known; the input is unclear. Repeating the same calculation cannot recover the missing digit.

For this paper investigation, treat the listed possibilities as the complete set allowed by the exercise. In real use, that assumption would itself need evidence. We will compare the model's predictions under each possibility; we are not changing Tower1 to accept an invented exact measurement.

3. See how it works

VISUAL MECHANISM

Input → **Process** → **Output**

Follow the information instead of imagining magic inside the machine.



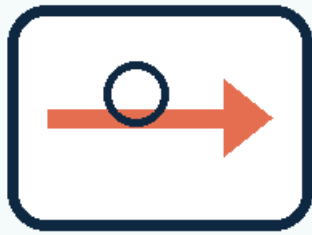
INPUT

features with incomplete or conflicting
clues



PROCESS

compare evidence with category
boundaries



OUTPUT

category, not sure, or request more information

INPUT features with incomplete or conflicting clues

PROCESS compare evidence with category boundaries

OUTPUT category, not sure, or request more information

CORE IDEA Overlapping categories create uncertainty; abstaining can be responsible.

UNDER THE HOOD Abstention is a decision policy: when confidence or evidence fails a threshold, the system defers. Good thresholds balance coverage against error cost and specify what additional information or human review follows.

Draw a fresh example with all three stages:

A closer explanation

Possible heights can change the prediction

Available record · Predictions to compare · What can be stated

Available record: Either 5 or 6 cm

Predictions to compare: 5 gives short; 6 gives tall

What can be stated: Category unresolved

Available record: Either 8 or 9 cm

Predictions to compare: Both give tall

What can be stated: Tall under both inputs

Available record: No height supplied

Predictions to compare: Possibilities not listed

What can be stated: Request a measurement

A shared prediction across the listed possibilities does not identify the exact input or prove the category correct. Different predictions show that the unknown input can change this model's result.

At either 5 or 6 cm, one possibility predicts short and the other tall. Record no single category yet and ask for the original record or a new measurement. Do not select the more convenient digit. Averaging the two readings would introduce a different input that neither possibility stated.

At either 8 or 9 cm, the model predicts tall in both cases. You can state that the prediction stays tall under these two possibilities. Keep the exact-height uncertainty visible. This agreement does not check the model against a correct task label or establish that the list of possible heights is complete.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

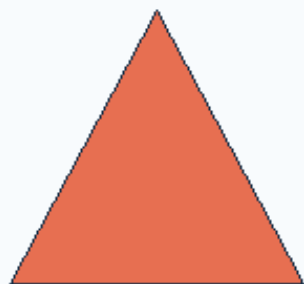
PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the evidence carefully.



Clearly a book



Toy ball



Covered object



Personal item

CASE 1 — Clearly a book Clue: strong single-category evidence. Meaning: classify.

CASE 2 — Toy ball Clue: toy and sports overlap. Meaning: review purpose.

CASE 3 — Covered object Clue: important features hidden. Meaning: need more info.

CASE 4 — Personal item Clue: high consequence if wrong. Meaning: adult review.

BOUNDARY Responsible uncertainty is informative and actionable, not a blank refusal.

Which two pictures are easiest to confuse? What evidence separates them?

Change one thing in the pictured case

Change: Narrow B's possibilities to 6 or 7 cm.

Prediction: Both fail the width limit; rejection no longer depends on resolving their difference.

Reason: All remaining possibilities lead to the same decision.

5. Words that unlock the idea

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.





uncertainty



abstain



Word · Meaning · My example

Word: boundary case

Meaning: example near competing category rules

My example: _____

Word: uncertainty

Meaning: limited support for one conclusion

My example: _____

Word: abstain

Meaning: decline to choose a category

My example: _____

Word: escalation

Meaning: send the case for added information or review

My example: _____

Misconception detective

CLAIM TO REPAIR A good system must always choose one label.

Use one vocabulary word and one piece of evidence to repair it:

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

Word · Meaning

Word: Possible input

Meaning: An input still compatible with the stated information

Word: Review trigger

Meaning: A stated condition that sends a case for an additional check

Word: Review evidence

Meaning: Information gathered to investigate the reason a case was referred

Word: Unresolved

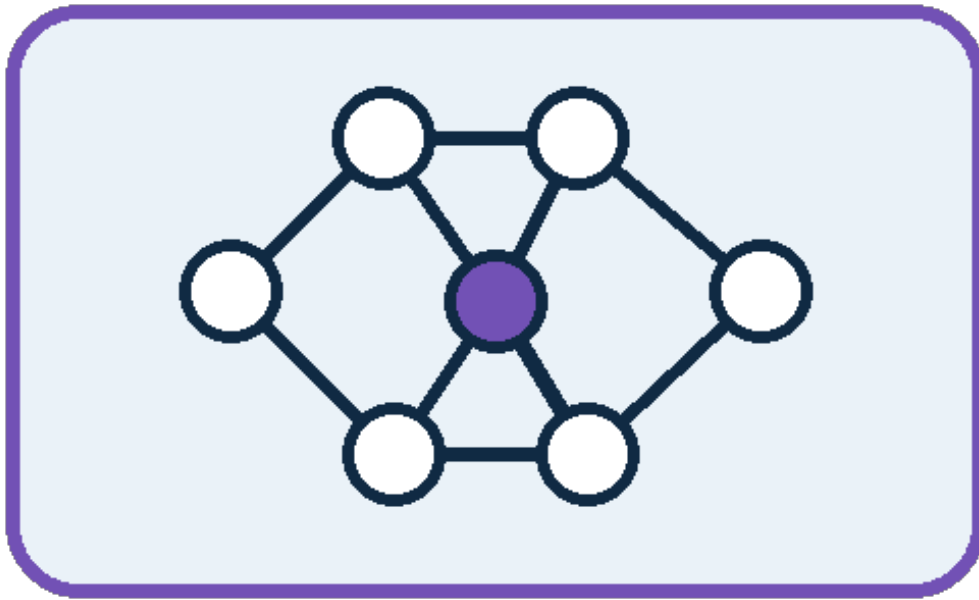
Meaning: Still not settled after the available checks

6. Follow a worked example

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable evidence.



CASE A soft toy shaped like a ball may fit toy, sports, or comfort-item bins depending on the sorter's purpose.

STEP 1 — OBSERVE features with incomplete or conflicting clues

STEP 2 — TRACE compare evidence with category boundaries

STEP 3 — CONCLUDE category, not sure, or request more information

MODEL REASONING A useful not-sure result is specific: it states which categories compete, what evidence is missing, and who or what should supply it. Abstention reduces forced errors but should not become an excuse to avoid all decisions.

DO NOT OVERCLAIM Responsible uncertainty is informative and actionable, not a blank refusal.

What new evidence could change this answer?

A closer explanation

Reason for review · Relevant check · What it cannot settle alone

Reason for review: Smudged height

Relevant check: Read the original or remeasure

What it cannot settle alone: Whether the model is accurate

Reason for review: Unit missing

Relevant check: Check how the record was made

What it cannot settle alone: The unit from digits alone

Reason for review: Two labels for one record

Relevant check: Check the label policy and sources

What it cannot settle alone: Truth by majority vote

Reason for review: Poorly tested new use

Relevant check: Evaluate relevant reserved cases

What it cannot settle alone: Every future outcome

A review request should carry the original input, model version, unresolved issue and any predictions already made. Name the evidence to seek and the person responsible for checking it. A reviewer needs access to relevant evidence and a clear task; being human does not make an unsupported guess correct.

After review, record what was checked and what changed. If a reliable new reading establishes 6 cm, the model predicts tall. If the record is still unreadable, keep no prediction with reason unresolved height. Do not replace the original record without retaining the reason for the correction.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Boundaries and 'Not Sure'

A fictional cubby selector accepts a block if width is at most 5 cm.

The facts we will use

An unclear measurement means Block A is either 4 or 5 cm; Block B is either 5 or 6 cm.

These complete possibility sets contain all widths under consideration. Other dimensions are known to fit.

Possibilities and decisions

	Width 1 cm	Decision 1	Width 2 cm	Decision 2
Block A	4	fits	5	fits
Block B	5	fits	6	fails

Read a cell using both its row and column labels.

Boundaries and 'Not Sure'

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Block A possibilities

4 and 5 both satisfy the including equality width limit.



2. Block B possibilities

5 fits; 6 does not.



3. Compare decisions

A has one supported decision; B's possible decisions disagree.



4. Review B

Ask a named reviewer to remeasure B using the specified width unit.

CHANGE ONE THING

Narrow B's possibilities to 6 or 7 cm.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Block A possibilities

What happens and why: 4 and 5 both satisfy the including equality width limit.

Step: 2. Block B possibilities

What happens and why: 5 fits; 6 does not.

Step: 3. Compare decisions

What happens and why: A has one supported decision; B's possible decisions disagree.

Step: 4. Review B

What happens and why: Ask a named reviewer to remeasure B using the specified width unit.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Uncertainty about a number need not always create uncertainty about the decision. Every allowed width for A fits, so its width uncertainty does not affect acceptance. B straddles the boundary. Record both possibilities and request a better measurement instead of selecting the one that makes a preferred outcome.

What this example does not establish

A review path is not automatically accurate; its conclusion depends on relevant, better evidence.

7. Find and repair a mistake

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case · Evidence · What it shows

Case: Clearly a book

Evidence: strong single-category evidence

What it shows: classify

Case: Toy ball

Evidence: toy and sports overlap

What it shows: review purpose

Case: Covered object

Evidence: important features hidden

What it shows: need more info

Case: Personal item

Evidence: high consequence if wrong

What it shows: adult review

Repair two tempting claims

CLAIM A A good system must always choose one label.

Repair:

CLAIM B Not sure means the system failed completely.

Repair:

STRONG ANSWER FRAME I claim _____ because the evidence shows _____. The process is _____. One limit is _____.

A closer explanation

A deterministic model repeats the same prediction for the same input and settings. Ten identical runs do not give ten independent measurements of a smudged digit. Several models can also share a feature gap, training examples or mistaken assumption. Agreement alone is not proof that the prediction is correct.

Keep the claim specific: tall under both listed heights, input still unclear, or no category until the measurement is checked. Do not attach a made-up confidence percentage. Evidence about correctness and the meaning of confidence will be developed in Weeks 17 and 18.

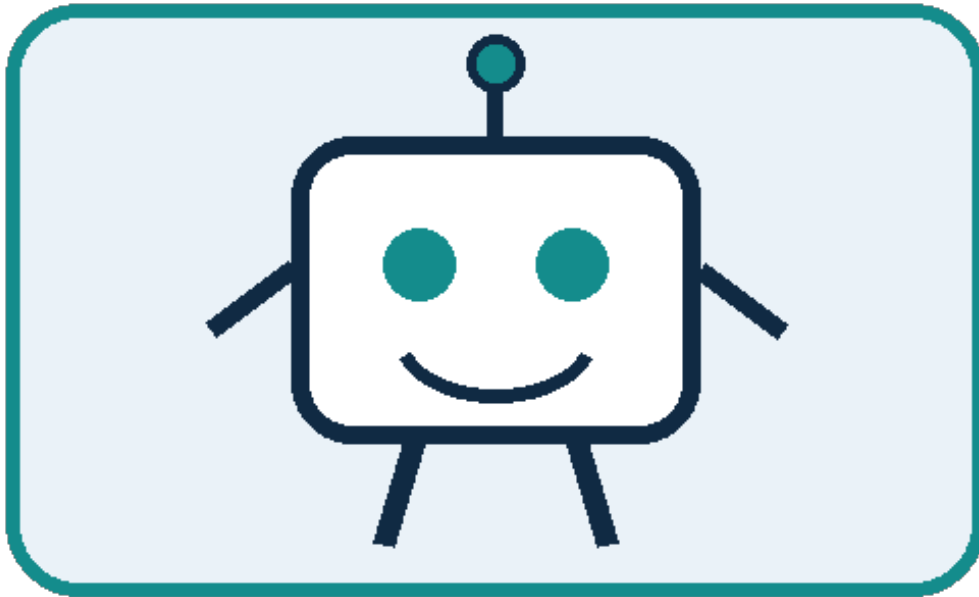
A learner runs the same model ten times on a guessed height and gets tall every time. They say review is no longer needed. Explain what the repetitions establish and name the evidence still needed

8. Practise together

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: Overlapping categories create uncertainty; abstaining can be responsible.

Use a fresh example to explain boundary case.

Trace a new input → process → output example.

Find and repair the error in: Not sure means the system failed completely.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the evidence. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Use a written review policy: if stated possible valid heights lead to different predictions, pause the single-category output and request a measurement check. If a required input or its unit is missing, request that information. After the check, apply the model only to a supported valid input; otherwise retain an unresolved outcome.

A card could mean 5 or 6 cm. Write the record sent for review, the check requested and two possible endings: a new measurement establishes 6 cm, or no reliable height can be obtained. Explain what each ending permits you to report

A review path needs an ending even when evidence cannot be obtained. Repeating the same unsuccessful request forever does not resolve the uncertainty. Record that the case remains unresolved and follow the agreed next step, such as holding the classroom sorting decision until the missing record is supplied.

Practise with less help: Keep a stable prediction within its limits

Use workbook W2 for the connected example “Uncertainty and review”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Uncertainty and review. Proposed explanation: “Two possible values imply equal probabilities”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Which original block needs a better width before accepting or rejecting it? What should the reviewer do?

My explanation using a pictured step and the given facts

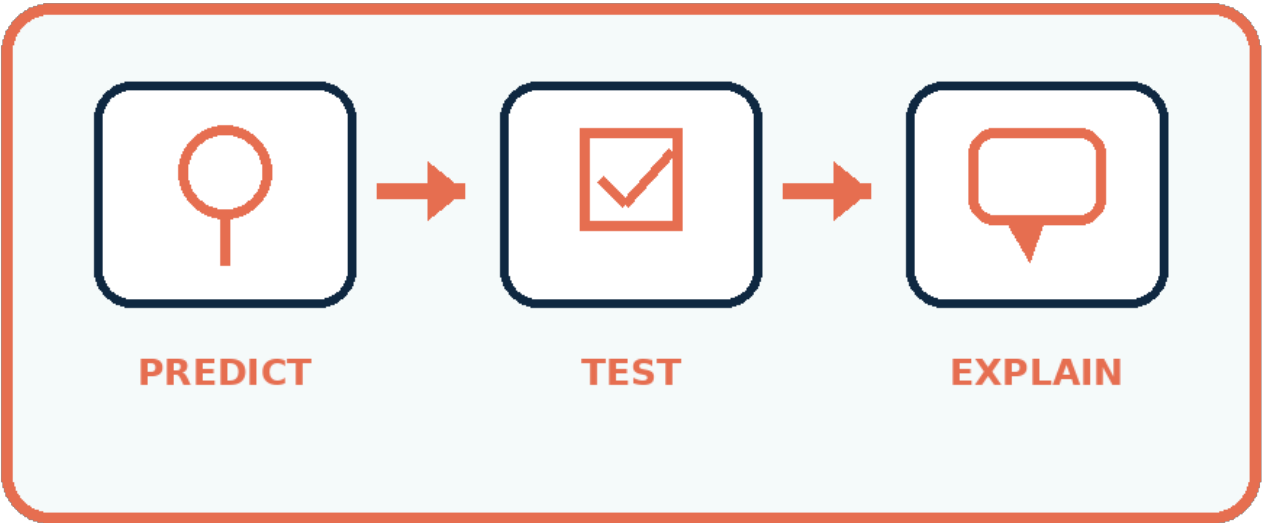
This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EXPLORER LAB

Boundary and Abstention Studio

Predict first, test honestly, and explain what the evidence means.



State your prediction before the result is known.

Complete the task: Find three boundary cases and design a 'need more info' output.

Keep a normal case and an edge case clearly separated.

Record what happened—even when it does not match your prediction.

Explain the result, one limitation, and the next useful test.

Case · Prediction · Observed result · What it means

Case: Normal

Prediction: _____

Observed result: _____

What it means: _____

Case: Edge

Prediction: _____

Observed result: _____

What it means: _____

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

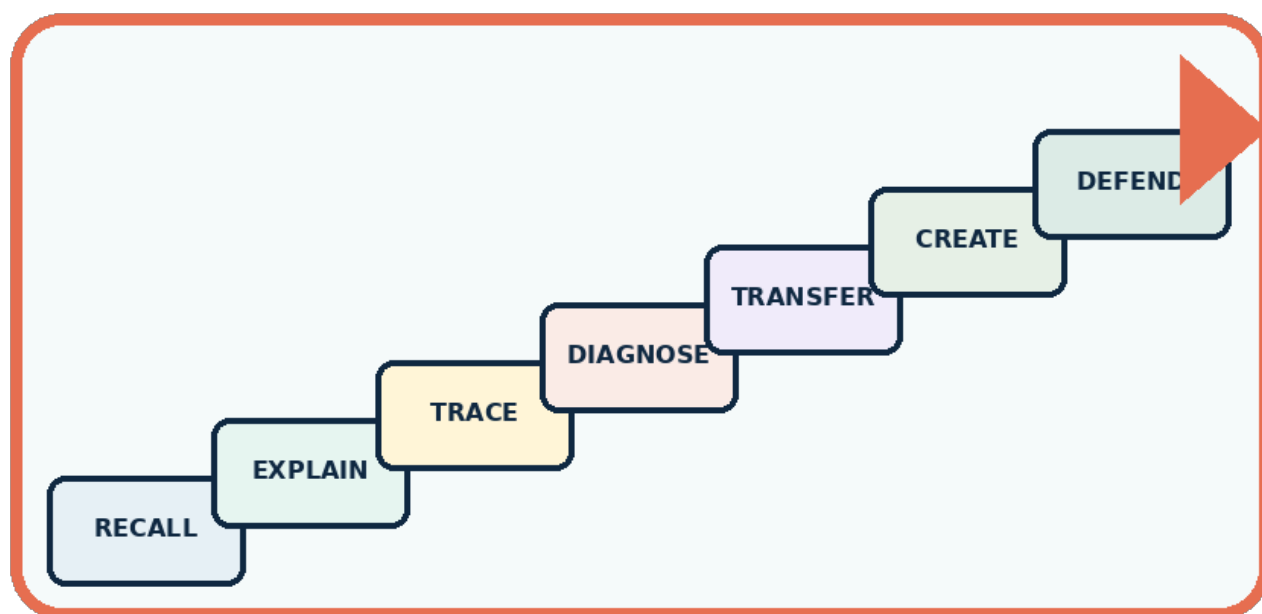
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RECALL Define boundary case.

EXPLAIN Overlapping categories create uncertainty; abstaining can be responsible.

TRACE Show the input, process, and output.

DIAGNOSE Repair: A good system must always choose one label.

TRANSFER Apply the idea to a new home or classroom case.

CREATE Design one revealing test.

DEFEND Give a claim, evidence, mechanism, limit, and safe action.

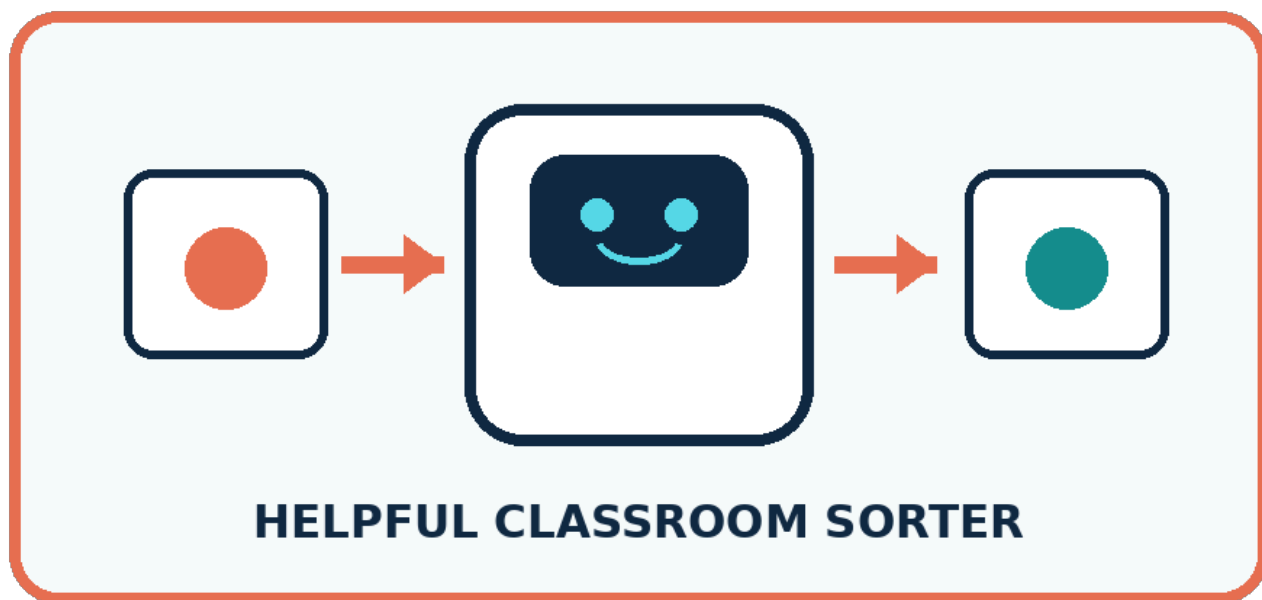
RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

11. Connect your project

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one evidence-based decision at a time.



THIS WEEK'S CHECKPOINT Add a not-sure branch, missing-information prompt, and human review destination to the sorter.

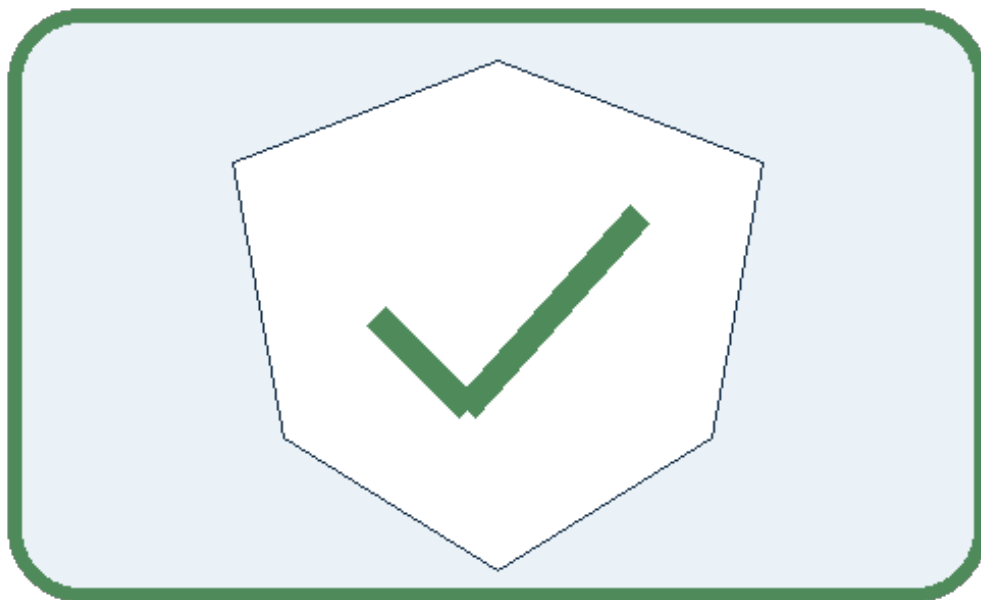
What real classroom need does this serve?

What information is necessary—and what should stay private?

What happens inside the process?

What normal and edge cases will you test?

When should it say not sure or ask a person?



RESPONSIBLE DESIGN Responsible uncertainty is informative and actionable, not a blank refusal.

My project decision:

Evidence for your project

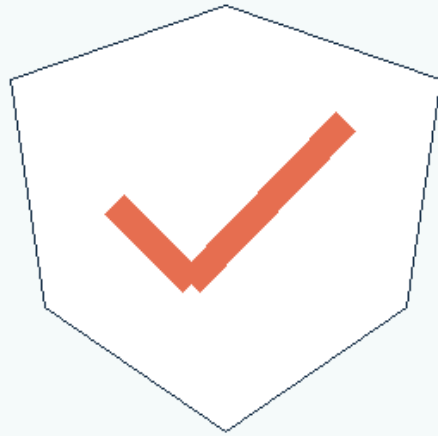
Add explicit review paths to the sorter, including unresolved input, responsible person and evidence needed.

12. Show what you know

MASTERY

Fresh-Case Independent Proof

Complete this after teaching and guided practice have ended.



INDEPENDENT PROOF

REQUIRED PROOF Find three boundary cases and design a not-sure output that names the missing clue and next action.

- ☐ Name or draw the fresh case.
- ☐ Trace input, process, and output.
- ☐ State the conclusion and strongest evidence.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Independent evidence:

SELF-CHECK A useful not-sure result is specific: it states which categories compete, what evidence is missing, and who or what should supply it. Abstention reduces forced errors but should not become an excuse to avoid all decisions.

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

NEXT BRIDGE Carry today's evidence habit into: Prediction Is Not Fact.

13. Study a complete case

Week 16 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This

case extends the illustrated lesson.

The situation

A sorter handles clearly red and clearly blue cards. Purple and very dark cards arrive.

The question

How should it handle each uncertain case?

Worked explanation

Use a review tray. Purple is outside the clear categories; darkness prevents observation. These are different uncertainty causes.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Two different reasons to ask for help

Evidence or stage · Value or result

Evidence or stage: Purple card

Value or result: Outside clear red/blue categories

Evidence or stage: Very dark card

Value or result: Colour cannot be observed

Evidence or stage: Review tray

Value or result: Preserve uncertainty for a person

A sorter handles clearly red and clearly blue cards. Purple and very dark cards arrive.

How should it handle each uncertain case?

Use a review tray. Purple is outside the clear categories; darkness prevents observation. These are different uncertainty causes.

14. Try a new case independently

Independent case: Transfer the method to distance

A paper model predicts far at a distance of at least 4 units, otherwise near, for whole-number distances 1 through 8. For each complete possibility set, compare predictions: 2 or 3; 3 or 4; 4 or 5. Identify the set requiring review before a single category can be stated. Design a relevant check and an unresolved ending. Limit the claims for the other two sets.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 16 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A sorter knows pencils and books. You show a lunch box. Should it force a known label or ask for help? State a useful fallback.

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Add a not-sure branch, missing-information prompt, and human review destination to the sorter.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

A sorter receives a clear purple card, a red card hidden by a sleeve and a card labelled both red and blue

by two reviewers. Give a different reason for uncertainty in each case and a suitable next action.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.